My First Maze (Junior Edition): Teacher Guide

At a Glance

Age	8 to 9, using the **Junior** edition of PyGameMaker
Time	90 minutes
Students make	A one-screen maze: draw a hero and a goal, move the hero with the arrow keys, stop at walls, win at the goal, export as one .html file
Needs	PyGameMaker in Junior mode, the starter project, a web browser. Internet is **not** needed to play this game
Student sheet	Junior Maze: Student Handout (one for each student)

Everything in the lesson uses only blocks that exist in the Junior toolbox. The finished game was run in the real game engine, exported, and played in a browser to check it.

Before the Lesson (15 minutes, once)

1. **Switch on Junior mode** on each computer: Tools > Preferences > General > IDE Edition > Junior, then restart the IDE.
2. **Copy the starter project** (Junior-Maze-Starter.zip, from the wiki page or wiki/downloads/) and unzip it so each student has a folder called **My Maze**.
3. **Try the lesson yourself once.** The finished version is in Junior-Maze-Finished.zip to compare with.
4. **Decide how the game will get to a phone** (see "Playing on a Phone" below). Do this before class, not after.
5. Print the student handout, or put it on a second screen.

> If a computer was ever set to another theme or font in Preferences, Junior's bigger text will not apply. Reset them there.

What the Starter Contains

A brown wall picture (spr_wall), a **Solid** wall object (obj_wall) and an empty 640 by 416 room (room0) with its border already built. Students draw their own hero and goal and do the rest. The room is on a 32-pixel grid and snap-to-grid is on.

Window size. The game window is exactly the size of the room, which is small on purpose here (640 by 416 pixels, 20 by 13 squares) so the whole maze can be built and checked in one lesson. If a student wants a bigger playing area, do not stretch the room (the border walls would be left open on the new edge); point them to the next lesson, "Treasure Maze", whose starter room is 960 by 640 pixels.

Lesson Plan (90 minutes)

Time	Part	What happens	If you are late
0-10	Warm-up	Show the finished game (play it, ideally on a phone). Ask: what does it need? A hero, walls, a goal	Skip the discussion
10-15	1. Open	Open the starter, press F5	
15-30	2. Sprites	Create Sprite twice (hero, goal)	Give them the art from the Finished project
30-50	3. Hero	Create Object, six Blockly rows	Show the first row on the board, then they copy it
50-60	4. Goal	One object, two blocks	
60-80	5. Maze	Paint walls, place hero and goal, test	Give a ready maze: open the Finished room
80-88	6. Export	Export as HTML5, open in browser	Do the export as a class demo
88-90	Wrap-up	Swap and play; the "Think About It" questions	

Parts 2 to 4 are the same for everyone. Part 5 is where students spend their creativity, so protect that time.

Walkthrough and Common Problems

Part 2, drawing. Create Sprite opens a blank 32 by 32 picture in the sprite editor. Remind students to click Save in the editor. A sprite name that exists already gives a "Name Already Used" message instead of overwriting.

Part 3, the hero. Six events: four held arrow keys, No key, and a collision with obj_wall. The rows are copy-able: students use right-click > Duplicate and change the key and number.

- Without **No key** the hero never stops.
- Without the **collision** row the hero walks straight through the walls: the wall being Solid does not stop it by itself (checked: it left the room). The collision row with **Stop movement** is what makes the wall a wall.
- A speed of 4 moves the hero 4 pixels each step, 60 steps a second.

Part 4, the goal. The goal object holds the win logic: *when colliding with obj_hero: Show message, then Restart room*. The message can say anything, and punctuation is safe.

Part 5, the maze.

- **Two squares wide.** This is the rule that matters most. The hero stops wherever a key is released, not on the grid. In a one-square path it can end up a few pixels off and jam. Every corridor in the reference maze is two squares wide, and a test plays it with the hero placed up to 24 pixels off the grid and always reaches the goal.
- Holding the mouse down and dragging paints a row of walls. A click places one.
- Check that the maze can be finished before swapping.
- Place the hero and the goal inside the border, on empty squares.

Exporting

File > Export as HTML5 asks for a folder and offers to open the game in the browser. The result is **one file**, named after the project. This game has no Python code, so it needs no internet to play. The window above the game shows **Score** and **Lives**; this maze does not use them, so students may ask about them. They can be ignored.

Playing on a Phone

What is known:

- The exported page detects a touchscreen and then shows four arrow buttons under the game. Holding a button moves the hero. This was checked in a phone-sized browser on a computer, **not on a real phone**.
- The game page is built for narrow screens and shrinks to fit.

What to plan for:

- **Getting the file onto the phone.** Phones often cannot open an .html file from their storage (iPhones in particular). The dependable way is to put the file on a **web address** (the school's web space or a free site host), then open that address on the phone. This was not tested here.
- **Speed on a fast screen.** The exported game takes one step per screen refresh. On a 60 Hz screen this is the intended speed. On a phone with a 90 or 120 Hz screen the hero will move faster, up to twice as fast. This is a known limit of the HTML5 export that has not been fixed. A maze is still playable, but keep it in mind for any game that needs careful timing.

Differentiation

- **Needs support:** start from the Finished project's room and only change the walls, or give a printed maze to copy.
- **Going further:** add coins that give points (Tutorial 2, "Your First Game", teaches the score blocks), or a second goal. Both use blocks that are in the Junior toolbox.

Assessment

Ask each student to show you: a sprite they drew, a hero that moves and stops at walls, a maze that can be finished, and the exported .html file working in a browser. The three "Think About It" questions work as an exit ticket.

Troubleshooting

Symptom	Cause and fix
Hero does not move	The hero is not placed in the room, or the arrow rows are missing
Hero goes through walls	Missing the collision row (**When colliding with** obj_wall, **Stop movement**) on the hero
Hero stuck in a path	Path is one square wide. Widen it

Nothing at the goal	Goal needs the collision event with obj_hero
"You win!" repeats	It should show once, then the room restarts. Check **Restart room** is under the same event
Grey game window	Slow or low-memory computer. Wait, then try again
Something deleted	Tools > Restore Deleted Assets